Evaluate yourself – 6

a) Answer

$$\underset{10^{-8}\ M}{H_2SO_4} \xrightarrow{H_2O} \underset{2\times10^{-8}}{2H_3O^+} + \underset{10^{-8}\ M}{SO_4^{2-}}$$

In this case the concentration of H_2SO_4 is very low and hence $[H_3O^+]$ from water cannot be neglected

$\therefore [H_3O^+] = 2\times10^{-8}$ (from H_2SO_4) + 10^{-7} (from water)

$=10^{-8}(2+10)$

$=12\times10^{-8} = 1.2\times10^{-7}$

$pH = -\log_{10}[H_3O^+]$

$= -\log_{10}(1.2\times10^{-7})$

$=7-\log_{10}1.2$

$=7-0.0791=6.9209$

b) Answer

pH of the solution = 5.4

$[H_3O^+]$ = antilog of (-pH)

= anitlog of (-5.4)

= antilog of $(-6+0.6) = \bar{6}.6$

$= 3.981\times10^{-6}$

i.e., 3.98×10^{-6} mol dm^{-3}

c) Answer

No of moles of HCl $= 0.2\times50\times10^{-3} = 10\times10^{-3}$

No of moles of NaOH $= 0.1\times50\times10^{-3} = 5\times10^{-3}$

No of moles of HCl after mixing $= 10\times10^{-3} - 5\times10^{-3}$

$= 5\times10^{-3}$

after mixing total volume = 100mL

$\therefore$ Concentration of HCl in moles per litre = $\frac{5\times10^{-3}\text{mole}}{100\times10^{-3}\text{L}}$

$[H_3O^+]=5\times10^{-2}$ M

$pH = -\log(5\times10^{-2})$

$= 2-\log 5$

$= 2-0.6990$

$= 1.30$

Evaluate yourself - 7

$$\alpha=\sqrt{\frac{K_b}{C}}=\sqrt{\frac{1.8\times10^{-5}}{6\times10^{-2}}}$$

$=\sqrt{3\times10^{-4}}$

$=1.732\times10^{-2}$

$=\frac{1.732}{100}=1.732\%$

Evaluate yourself – 8

a) Answer

Dissociation of buffer components

$$NH_4OH\ (aq) \rightleftharpoons NH_4^+(aq)+OH^-(aq)$$

$$NH_4Cl \rightarrow NH_4^+ + Cl^+$$

Addition of H+

The added H^+ ions are neutralized by NH_4OH and there is no appreciable decrease in pH.

$$NH_4OH(aq) + H^+ \rightarrow NH_4^+(aq) + H_2O(l)$$

Addition of OH^-

$$NH_4^+(aq) + OH^-(aq) \rightarrow NH_4OH\ (aq)$$

The added OH^- ions react with NH_4^+ to produce unionized NH_4OH. Since NH_4OH is a weak base, there is no appreciable increase in pH

b) Answer

pH of buffer

$$\underset{0.4-\alpha}{CH_3COOH(aq)} \rightleftharpoons \underset{\alpha}{CH_3COO^-(aq)} + \underset{\alpha}{H^+(aq)}$$

$$\underset{0.4}{CH_3COONa(aq)} \longrightarrow \underset{0.4}{CH_3COO^-(aq)} + \underset{0.4}{Na^+(aq)}$$

$$[H^+]=\frac{K_a[CH_3COOH]}{[CH_3COO^-]}$$

$[CH_3COOH]=0.4-\alpha \simeq 0.4$

$[CH_3COO^-]=0.4+\alpha \simeq 0.4$

$$\therefore\ [H^+]=\frac{K_a(0.4)}{(0.4)}$$

$[H^+]=1.8\times10^{-5}$

$\therefore\ pH = -\log(1.8\times10^{-5}) = 4.74$

Addition of 0.01 mol HCl to 500ml of buffer

$$\text{Added } [H^+]=\frac{0.01\text{ mol}}{500\text{ mL}}=\frac{0.01\text{ mol}}{\frac{1}{2}\text{L}}$$

$= 0.02M$

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